RAUNAK DEY

IIT Madras
B.S. Degree

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SUMMARY

Data Science and Machine Learning student with experience in AI systems, information retrieval, quantitative research, and full-stack engineering. Builds and evaluates data-driven systems using machine learning, embeddings, retrieval pipelines, and statistical experimentation, with an interest in reliable AI and cloud-based systems.

EDUCATION

Indian Institute of Technology Madras — Chennai, Tamil Nadu
BS in Data Science and Applications | CGPA: 7.21/10 — *September 2022 – August 2027*

SKILLS

Languages: Python, SQL, TypeScript, JavaScript
Frameworks: FastAPI, Next.js, React, Flask
AI/ML: PyTorch, Scikit-learn, Sentence Transformers
Data & Infrastructure: PostgreSQL, pgvector, Redis, Docker

EXPERIENCE

Quantitative Research Consultant — April 2024 – March 2025
WorldQuant — *Remote*

- Developed and evaluated 15 quantitative alphas across price-volume, news, options, and social-media datasets, using 25 unique operators and 28 financial data fields
- Completed 4 alpha pyramids spanning News, Options, Price-Volume, and Social Media datasets
- Achieved combined alpha performance of 0.82 and Power Pool performance of 1.12 through iterative signal research and evaluation

PROJECTS

Verilore | *Python, PostgreSQL, pgvector, ONNX Runtime, FastAPI, MCP* [GitHub] — August 2026 – Present

- Built an open-source, versioned memory engine for AI that records source-attributed claims with exact-quote evidence, temporal validity, cross-document supersession, and offline-verifiable SHA-256 receipts
- Enforced tenant isolation with PostgreSQL Row-Level Security across 13 tables and append-only triggers; diagnosed HNSW post-filtering starvation under RLS (0/5 neighbors for a 50-claim tenant sharing an index with 10,000 claims) and implemented a corpus-scoped B-tree bitmap scan fallback, restoring exact 5/5 recall in 5.6ms
- Migrated the default embedding pipeline to ONNX Runtime and tokenizers (216MB installation), verifying PyTorch equivalence at 0.99999833 minimum cosine similarity across truncation, non-ASCII, and code inputs; retained PyTorch as an optional cross-encoder reranking dependency and exposed CLI, Python SDK, FastAPI REST, and 17 MCP tools

Fin-alysis | *Next.js, TypeScript, Kubernetes, Helm, Terraform, Prometheus* — May 2025 – October 2026

- Built a financial research platform covering 2,300+ NSE equities, integrating fundamental analysis, price momentum, news sentiment, and explainable recommendations through resilient multi-source data pipelines
- Engineered fault-tolerant serverless APIs with rate limiting, circuit breakers, source fallbacks, and scheduled refresh workflows to handle upstream failures and financial data freshness
- Containerized the application and developed Kubernetes deployments, Helm charts, Terraform configuration, and Prometheus/Grafana observability, with automated Kubernetes CI and failure testing

CV-Embed | *React 19, TypeScript, Vite, Playwright, Zod* — February 2026 – October 2026

- Built an embeddable resume platform with a versioned, bidirectional postMessage SDK, shared validation schemas, and secure iframe communication for integration into external portals
- Engineered structured resume editing, real-time previews, and PDF, DOCX, and JSON export pipelines; optimized the SDK to 20.40kB gzip with 81ms builder first paint and 62ms preview updates
- Developed browser regression coverage across Chromium and Firefox with 138 unit tests and 165 passing E2E tests, uncovering and fixing 3 previously undetected SDK defects

ACCOMPLISHMENTS

Kaggle August 2025 – Present

Notebooks Master | Rank 75/60,358

- Earned **3 Gold, 8 Silver, and 8 Bronze medals** across 20 completed competitions, including 22 solo competition entries
- **BirdCLEF+ 2026:** Ranked 795 out of 4,094 teams by developing an ensemble pipeline combining ProtoSSM and BirdNET models
- **NVIDIA Nemotron Model Reasoning Challenge:** Ranked 1,715 out of 4,182 teams in an LLM open-model reasoning benchmark

International Quant Championship 2024 March 2024 – June 2024

WorldQuant

- Ranked 165 globally in Stage 1 and 184 globally in Stage 2 among thousands of participants
- Contributed to a team OS Score of 36,000+ using 24+ financial data fields for quantitative signal generation
- Collaborated in a two-member team to design and implement alpha strategies under competition constraints

CERTIFICATIONS

OCI Certified Foundations Associate
Credential ID: 331754025OCI26FNDCFA

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Credential ID: 331754025OCI26AICFA

EXTRA-CURRICULAR ACTIVITIES

Deputy Secretary March 2024 – August 2025
Diplomacia, the IITM Soft Skills Society *SA Contribution Certificate*

Marketing Team Volunteer August 2024 – December 2024
Student Placement Council, IITM BS Degree *SPC Impact Report*